

Let Us C — Chapter 1: Getting Started

Classic Programming Foundation • Dennis Ritchie Model • JUIT Scholar Study Material

1. Four Building Blocks of Computer Languages

- The way it stores data in computer memory (Types, Variables, Allocation).
- The way it operates upon this data (Arithmetic, Relational & Bitwise Operators).
- How it accomplishes Input and Output (scanf, printf, file streams).
- How it lets you control the sequence of execution of instructions (Control Flow & Loops).

Many believe nobody can learn C++ or Java directly without first having a solid grasp of core fundamentals.

Learning complicated OOP concepts (inheritance, polymorphism, exception handling) while still struggling with loops and pointers is putting the cart before the horse.

2. Why Learn C Before C++ or Java?

C provides raw, direct access to memory, hardware architecture, and execution mechanics.

The C Character Set & Memory Variables

1. Valid C Characters

- Alphabets: Uppercase A – Z, Lowercase a – z
- Digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

• Special Symbols: ~ ` ! @ # % ^ & * () _ + = | \ { } [] : ; " ' < > , . ? /

A constant is an entity that does not change during program execution.

A variable is an entity whose value may change.

Like human memory, computer memory consists of millions of storage cells. Each cell has an address, and giving a friendly name to a memory cell creates a variable name (e.g. `int x = 5;`).

2. Constants vs Variables